

Learning Forward and Inverse Kinematics for Soft Pneumatic Actuators for Control and Visualization of the RBO Hand 3

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Abstract—The RBO hand 3 is the latest soft robotic hand at TU Berlin. It is a highly compliant hand designed for dexterous grasping. The inherent compliance of the fingers makes it hard to model them analytically unlike their rigid links counterparts. Therefore, we propose an efficient data-driven approach to learn the forward and inverse kinematic models of the hand to control and visualize the hand in free space.

I. INTRODUCTION

The control of robotics manipulators has always been fascinating and challenging. There exist well-defined methods for finding the forward and inverse kinematics of rigid link manipulators. These methods exploit the geometry of rigid links, for example - transformations from one link to another are easily computed using DH parameters. Whereas on the other side of the spectrum, in the domain of soft-robotics, the RBO hand 3's fingers are soft silicon-based continuum actuators with two air chambers each. The fingers undergo a large number of elastic deformations and exhibit very high mechanical degrees of freedom. Therefore, the inherent compliance makes it hard to model the RBO hand 3 analytically unlike the rigid link manipulators to study the forward and inverse kinematics. But some degrees of freedom such as the actuators on the palm of RBO hand 3 can be approximated as rigid revolute joints because of their construction and behaviour (Fig. 1).

In this paper, we exploit machine learning techniques such as regression and artificial neural networks to learn a model of the compliant behaviour of the fingers. The compliance motion of fingers can then be coupled with the rigid joint motion of the palm to learn the forward and inverse kinematics of the RBO hand 3 in a data-efficient manner.

The objective is to control and visualize the free motion position of the hand. The possible interactions/contact positions with the external objects will not be captured since controlling these will not be possible due to the complexity of the deformation. And this is not a problem, as beautifully stated by Raphael Deimel [1]:

“To fully utilize the advantages of mechanical compliance in soft robotics in general and soft pneumatic hands in particular, their control should focus on putting compliance to work, rather than counteracting it by controlling either position or force, as is customarily done for both hard and soft hands.”

II. RELATED WORK

Controllers based on kinematic models for soft robotic arms have been implemented in the past. Analytical models

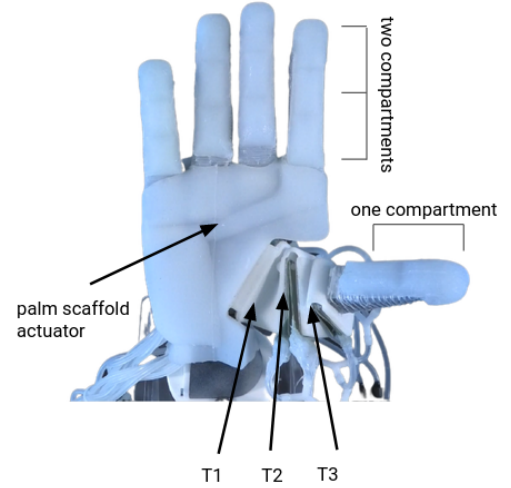


Fig. 1: RBO hand 3 has four fingers with two compartments (air chambers) and thumb is the only finger with one compartment. There are three actuators on the palm controlling the thumb joint motion - T1,T2,T3. The palm scaffold actuator on the middle of the palm is responsible for tilt motion of the palm, and therefore partially controlling the motion of ring and little finger.

have been established with simplified assumptions. A lot of the work is based on the simple assumption of piecewise constant curvature (PCC) [2]. The more complex analytical models are based on Beam theory and finite element methods (FEMs) [3]. However, these methods have a large computational cost in controlling the soft robots. Neural network methods, as they are good at learning non-linearities, can be used to model soft robots accurately [4][5]. However, neural network methods need a large amount of data to perform well owing to the non-linear characteristics of soft robots. Therefore, combining learning methods and analytical methods seems a lucrative option and can alleviate the problem of a huge data collection process[6]. However, all these methods don't take the effects of the environment into account and consequently, models will deviate while interacting with the environment due to the inherent compliance in soft robots.

For our soft robotic hand, combining the neural network method and analytical method is the best efficient way to control the hand owing to enormous data collection costs. However, the analytical method in our project is used to model parts of soft hand which behaves as rigid links.

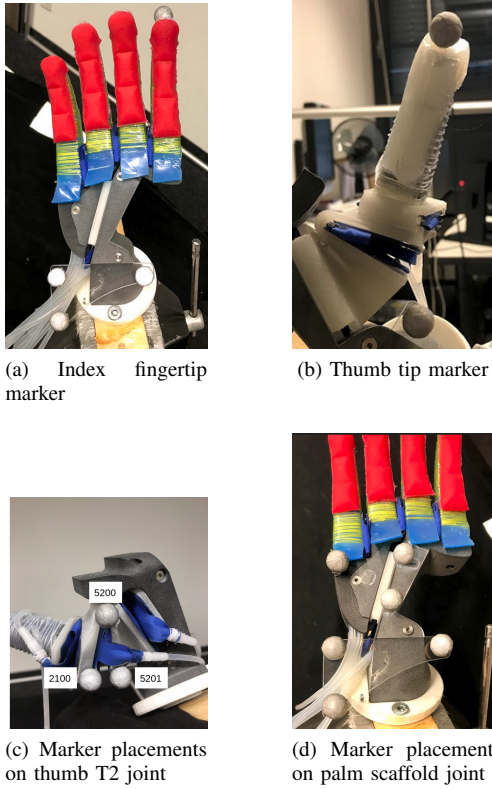


Fig. 2: Tracking finger tips and palm joint position w.r.t airmass levels.

III. TECHNICAL SECTION

A. MoCap data

To model the compliance behavior of the fingers, each fingertip position is recorded via a motion capture (MoCap) system for a series of inflation and deflation states (Fig. 2). Similarly, three key positions for each of the T1, T2, T3, and palm scaffold actuators are also traced. These key positions were chosen for MoCap marker placement in order to obtain the opening angle of the joint (Fig. 2, c & d).

Once the data is collected, it is visualized using python. Figure 3 shows the point clouds of all fingers tips for various cycles of inflation and deflation states. Note that, these workspaces only correspond to the compliance motion of the fingers i.e. no control exhibited on fingers by the palm actuators. The motion of the thumb tip in particular w.r.t airmass levels for the thumb compartment is shown in Fig. 4. The thumb tip starts at rest when the finger is deflated (zero airmass). Then with an increase in airmass levels (measured in mg), the thumb bends and the thumb tip exhibit the parabolic motion until it is fully inflated.

For the four actuators on the palm, the position of three markers is visualized. For one of the actuators (T2) on the palm controlling the thumb, the positions of these markers are shown in Fig. 5. This plot in itself doesn't provide meaningful information, and therefore, the data is preprocessed to get the joint angles w.r.t airmass levels as shown in Fig. 6. It

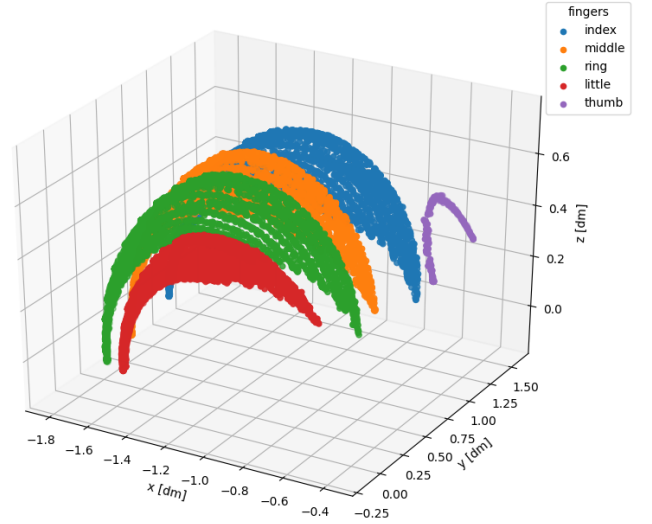


Fig. 3: Reachable workspace of all fingertips through compliance motion.

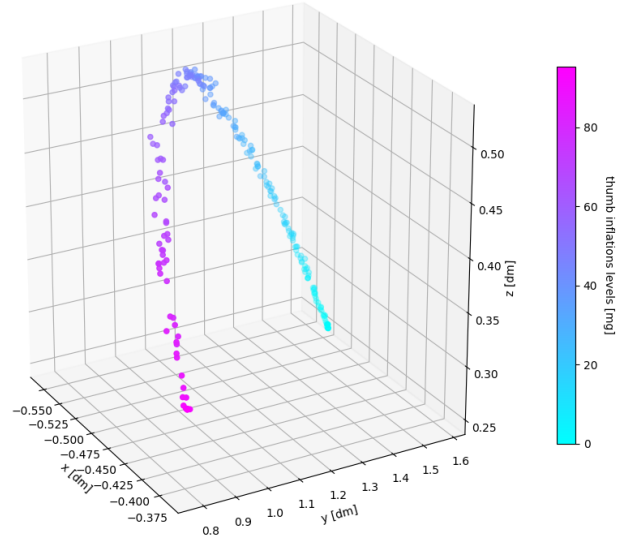


Fig. 4: Thumb compliance motion w.r.t airmass inflation levels.

was found that the maximum angle reached by the actuators T1, T2, T3 was significantly less than expected 90° . Also, the three positions that were traced on these actuators were not enough and reliable for accurate measurement on joint angles. Hence, it was decided not to consider and rely on these mappings between airmass levels and joint angles for the palm actuators (T1, T2, T3, and palm scaffold), instead to use joint angles for learning the kinematics.

B. Kinematic model

Using the CAD model of the RBO hand 3, the kinematics model is defined and also implemented in Rviz[7] for simulation. The DH parameters were crucial for defining the rigid

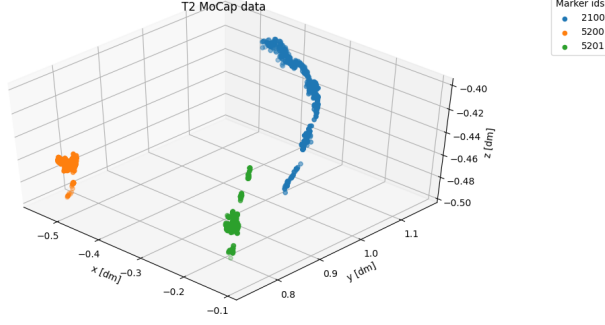


Fig. 5: Marker movements on T2 actuator for a series of inflation and deflation states. The markers 5201 and 5200 are static whereas the marker 2100 is the only moving marker. The opening angle of the t2 actuator thus can be calculated by forming two vectors - one originating from marker 5200 to 5201 and the other originating from marker 5200 to 2100. The angle between two vectors gives us the joint angle of t2 actuator which behaves as revolute joint.

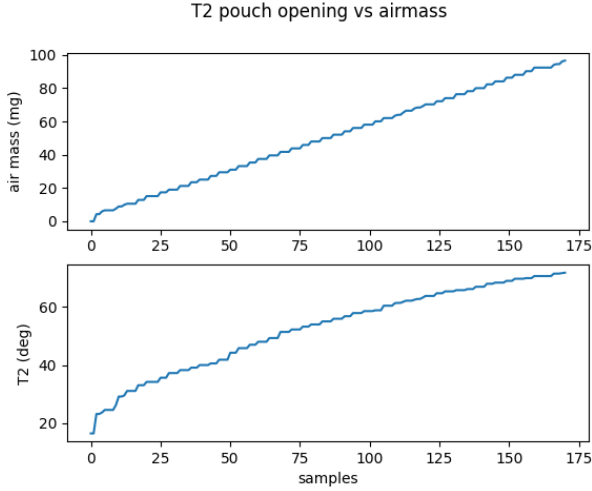


Fig. 6: Conversion of marker positions (Fig. 5) to joint angles for t2 actuator. The opening angle is directly proportional to the air mass levels and note that the maximum angle reached by t2 actuator is around 70° which ideally should be 90° . This reflects design problem in the pouch design of the pneumatic actuator.

links on the palm and were computed using the CAD model of the hand. The additional axes are placed on the base of the fingers namely index, middle, ring, little, and thumb as can be seen in Fig. 7. The compliant motion of the fingers is measured in these base frames and later coupled with other links to ultimately transform the fingertips positions into the palm frame.

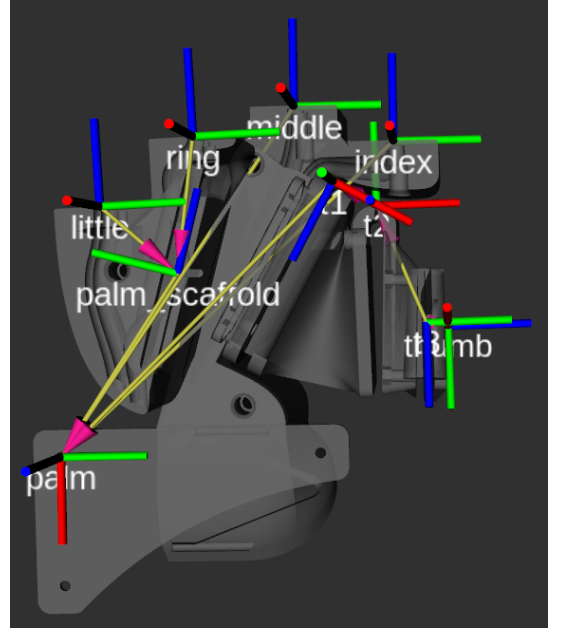


Fig. 7: Kinematic model of the RBO hand 3.

link i	$d(mm)$	θ°	$r(mm)$	α°
T1	0	θ_{T1}	0	0°
T2	-15.496	$30^\circ + \theta_{T2}$	10.370	-90°
T3	28.176	$-90^\circ + \theta_{T3}$	12.436	75°
thumb	5.0	180°	5.0	-90°

TABLE I: DH parameters for thumb kinematic chain.

C. Modeling compliance

As the finger undergoes a large number of elastic deformations it is hard to model it analytically due to large degrees of freedom. Therefore, we use neural networks to learn the non-linearities present in the compliant motion of the fingers. For the forward compliance motion model, the air mass levels of the fingers are mapped to 3D positions of the fingertips whereas the 3D positions are mapped to air mass levels for inverse compliance models. For each finger, the model is trained separately. The shallow neural networks proposed have hyperparameters shown in table II.

Property	Value	
	Dense (c_i)	Dense (c_i^{-1})
Number of layers	3	4
Number of units	30	20
Activation function	Tanh-Sigmoid	Tanh-Sigmoid
Optimiser	Adam	Adam
Loss	MSE	MSE

TABLE II: Hyperparameters for training neural network for compliance motion.

D. Index and middle finger

For this project, the motion of the index and middle finger solely relies on finger actuators i.e. the air compartments on

the finger itself and no dependence on the inflation state of the palm joint. Therefore, the forward and the inverse kinematics for the index and the middle fingers are simply given by equations 1 and 2, respectively.

$$\mathbf{X} = c_i(\boldsymbol{\mu}) \quad (1)$$

$$\boldsymbol{\mu} = c_i^{-1}(\mathbf{X}), \quad \forall i \in \{\text{index, middle}\} \quad (2)$$

where $\mathbf{X}$ is the position vector, $\boldsymbol{\mu}$ is the airmass vector, c_i and c_i^{-1} are the forward and inverse compliance model, respectively.

E. Ring and little finger

The forward kinematics of the ring and the little finger are modeled by coupling the compliant motion of fingers with the joint motion of the palm scaffold actuator. The fingertip positions are first computed in the base frame of the fingers (namely ring and little) using the compliance models. Then these positions are transformed into the palm frame of the hand taking into account the rotation of the palm scaffold joint using the transformation matrix as shown in Eq. 3.

$$\mathbf{X} = T(\theta_{\text{palm_scaffold}})c_i(\boldsymbol{\mu}), \quad \forall i \in \{\text{ring, little}\} \quad (3)$$

here, T is the transformation matrix from the base frame (ring, little) to the palm frame.

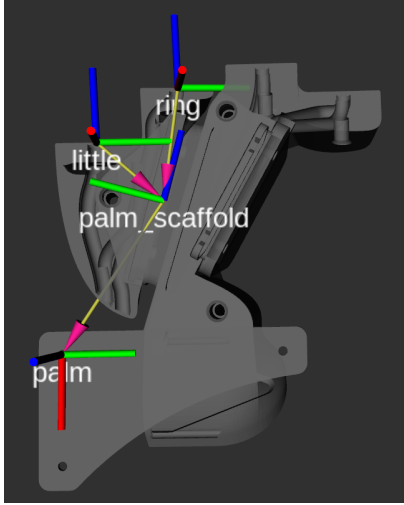


Fig. 8: Link transformations for little and ring finger.

To obtain an inverse kinematic model, we generate the samples from the entire workspace of these fingers using the forward kinematic model obtained in Eq. 3. The generated data is then fed into a shallow neural network to learn the inverse kinematics. The inverse mappings can be represented by Eq. 4.

$$\boldsymbol{\theta} = f_i^{-1}(\mathbf{X}), \quad \forall i \in \{\text{ring, little}\} \quad (4)$$

where $\mathbf{X}$ is the position vector, $\boldsymbol{\theta}$ is the configuration vector composed of airmass and joint values, and f_i^{-1} is the neural network model.

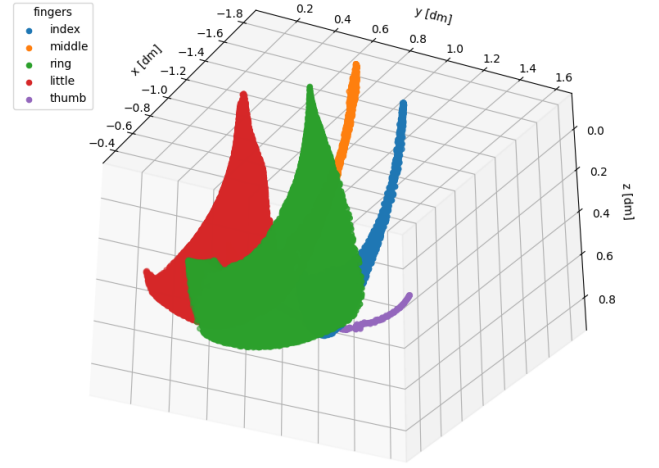


Fig. 9: Generated samples from entire workspace of ring and little finger.

Property	Value Dense (f_i^{-1})
Number of layers	4
Number of units	30
Activation function	ReLU-Sigmoid
Optimiser	Adam
Loss	MSE

TABLE III: Hyperparameters for training neural network for inverse kinematics of ring and little fingers.

F. Thumb

We follow a similar approach for the thumb as to the index and the little finger with the only difference of having more joints on the palm controlling the thumb. The forward kinematic model is given by Eq. 5, where the compliance motion in the *thumb* frame and the transformation of links (T_1, T_2, T_3) are coupled to get the thumb tip position in the palm frame of the hand.

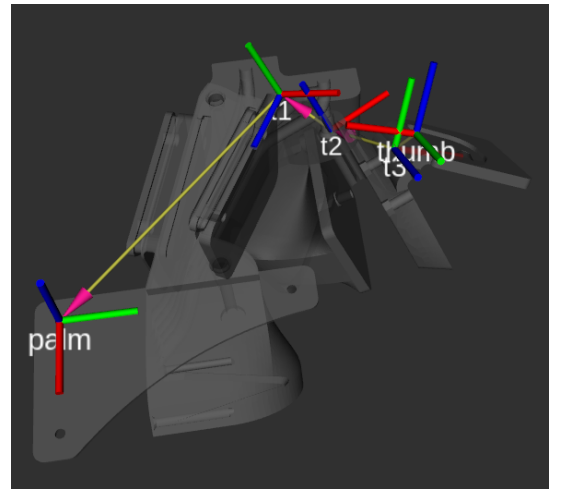


Fig. 10: Link transformations for thumb.

$$\mathbf{X} = T(\theta_{T1}, \theta_{T2}, \theta_{T3})c_{thumb}(\mu) \quad (5)$$

here, T is the transformation matrix from the base frame (thumb) to the palm frame.

To obtain the inverse kinematic model, samples from the entire workspace of the thumb are generated (Fig. 11) using the earlier obtained forward kinematic model. These samples are then learned using a KNN regression model. Neural networks were not utilized for the thumb due to resource and time constraints as the dataset contained approximately 76 million samples. The inverse mappings can be represented by Eq. 6.

$$\theta = f_{thumb}^{-1}(\mathbf{X}) \quad (6)$$

where $\mathbf{X}$ is the position vector, θ is the configuration vector composed of airmass and joint values, and f_{thumb}^{-1} is the KNN regression model.

Property	Value KNN
Number of neighbours	12
Number of data samples	76,110,671
weight function used in prediction	distance

TABLE IV: Hyperparameters for building the KNN regression model for learning the inverse kinematic model of the thumb.

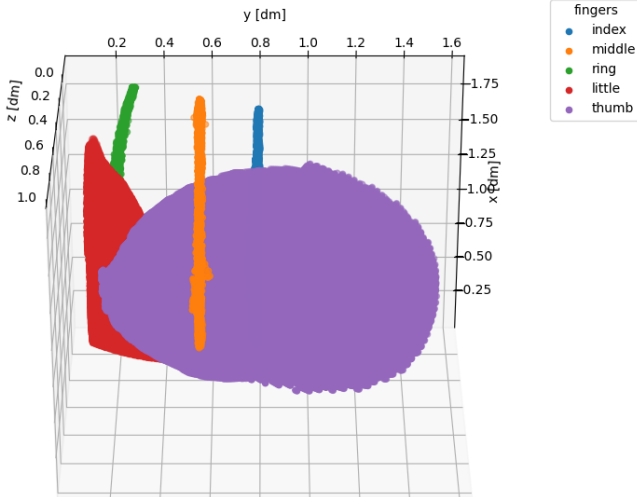


Fig. 11: Fully generated thumb workspace.

IV. EXPERIMENTAL RESULTS

The performance of the neural network based models for the forward compliant motion is stated in Table V and for the inverse kinematics is stated in Table VI.

Models	Loss RMSE
c_{index}	0.0557
c_{middle}	0.0417
c_{ring}	0.0729
c_{little}	0.0758
c_{thumb}	0.0611

TABLE V: RMSE of forward compliance models for all fingers.

Models	Loss RMSE
c_{index}^{-1}	0.0236
c_{middle}^{-1}	0.0318
f_{ring}^{-1}	0.0178
f_{little}^{-1}	0.0203
f_{thumb}^{-1}	0.1251

TABLE VI: RMSE of the inverse kinematic models. The inverse kinematic model of the thumb is based on KNN regression model and has a R^2 score of 0.82 on a prediction dataset of size 7.6 million. The performance of the model is good considering the complexity of the thumb.

A simulation environment was created in ROS and Rviz to demonstrate the capabilities of the RBO hand 3 based on the forward and the inverse kinematic models of the fingers.

In the first experiment, the compliant motion of the fingers is simulated for different levels of airmass in each of the fingers. The fingertip positions are computed using learned forward models and visualized in Rviz as shown in Fig. 12.

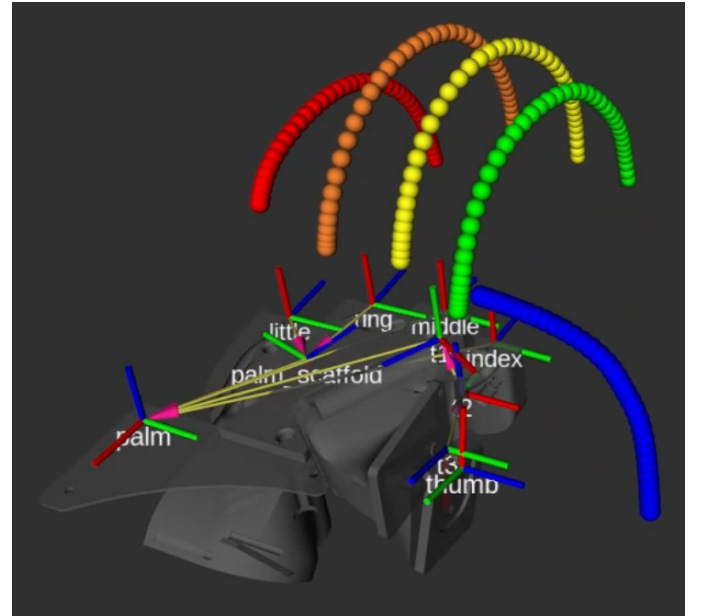


Fig. 12: Finger tips trajectories when fully inflated.

In the second experiment, the hand tries to perform few parts of the Kapandji test [8] in the simulation environment.

Kapandji test is used to measure the opposability of the thumb without measurement of angles. Usually, the test consists of touching the different parts of the four long fingers with the tip of the thumb. In our case, we limit the test to touching the four long fingertips only with the thumb tip because we don't model the entire fingers in our simulation. Through this test, we want to explicitly assess the opposability and the reachability of the thumb and implicitly, our obtained kinematic models.

For the thumb to touch the other fingertips, the airmass and the joint angle parameters are computed using learned inverse kinematic models. The thumb could easily oppose the other fingers by touching their tips as demonstrated in Fig. 13.

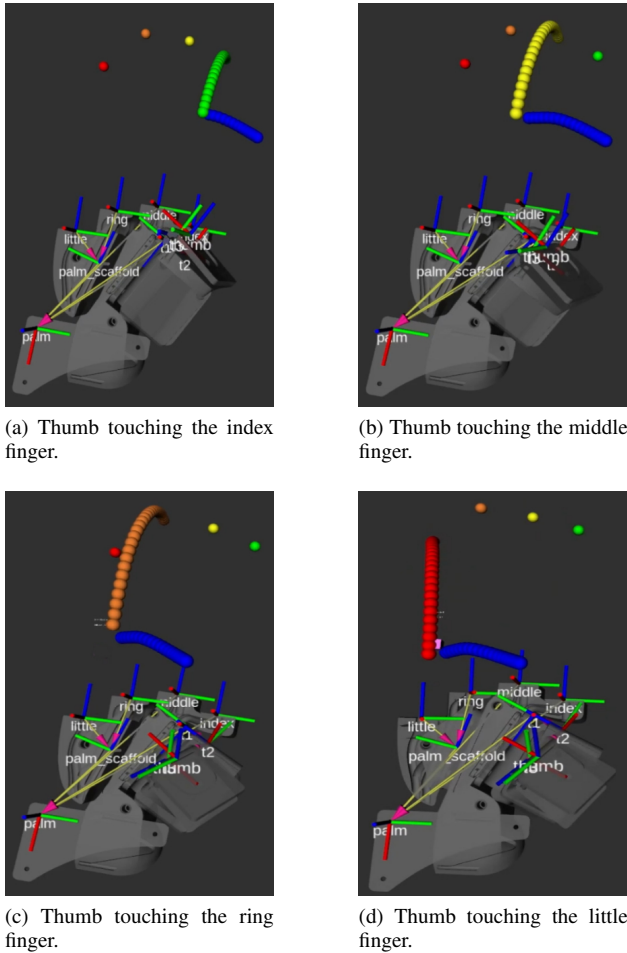


Fig. 13: RBO hand 3 performing the Kapandji test in the Rviz simulation environment.

V. CONCLUSION

The forward and inverse kinematic models of the fingers were successfully learned using machine learning techniques combined with the mechanics of the thumb and palm joints. Modeling the compliance motion independently and then coupling it with the joint motion make the learning data-efficient which otherwise would require capturing fingertips

position for all possible actuators configurations.

The simulation environment together with the kinematic modeling forms the modular framework useful for the following applications:

- 1) Grasping in the workspace: given the pose of the object to be grasped, the hand configuration can be computed using inverse kinematics.
- 2) Analysing and optimizing the workspace: changes can be made into the configuration of certain fingers such as the entire thumb kinematic chain, which can be modeled and simulated in the Rviz environment to access the opposability and reachability of the thumb even before building and 3D printing the real hand. Similarly, finger abduction motion can be modeled, learned, and simulated.

As part of the future work, we need to have a mapping between joint angles and airmass levels for palm actuators to be able to deploy these models on the real hand. Additionally, the RBO hand 3 shows drifts in the practical world, therefore the integration of airmass sensors into the control loop is needed for accurate deployment of the learned models. Lastly, the fingers can be visualized in the simulation environment with a chain of revolute joints assuming they maintain piecewise constant curvatures throughout the inflations.

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